

Communications research program

An original identity and supervised project menu for Mike, agents and prospective interns. This guide defines practical experiments, evidence standards and the decisions needed before launch.

MAKE THE SIGNAL USEFUL.

Industry Next Communications Lab explores how people, places and intelligent systems communicate under real constraints: weak connections, incomplete context, differing abilities and uncertain information.

2026 and 2027 explorations

Pay, hours, eligibility, supervisor, location, agreements and legal review must be resolved before recruitment or work begins. Budgets are illustrative materials estimates in USD, exclude labor, tax and shipping, and are not spending authorization.

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A small lab with clear evidence

The proposal takes inspiration from the breadth of industrial communications research. Nokia's history describes Bell Labs work on the transistor and information theory. Its current research overview emphasizes collaboration across mathematics, physics, computing and engineering. [1, 2]

Our interpretation is practical: combine disciplines around a testable question, then show the work. This is an original operating proposal, not a reconstruction of Bell Labs policies.

Start with a useful question

Name the person or situation the experiment could help. State what observation would change your mind.

Build the smallest revealing experiment

Use a baseline, synthetic fixtures and repeated trials before adding infrastructure.

Make uncertainty visible

Report failures, sample size, units, provenance and the limits of the evidence.

Keep authority human

Agents can draft, test and summarize. People approve participation, purchases, publication and consequential actions.

Leave a reproducible record

Archive code, methods and shareable evidence so another person can check the result.

An original signal language

The stepped signal and orange endpoint suggest a message becoming legible. The mark is deliberately technical and simple; it does not reproduce Bell Labs or telephone-company imagery.

**COBALT**

#1747D1

**INK**

#171B22

**ORANGE**

#E65B2A

Typography and layout

Bold system sans-serif for headlines; regular system sans-serif for reading; monospace for IDs and measurements. Use an 8 px web spacing rhythm, generous margins and a clear distinction between evidence and interpretation.

Use and accessibility

Use cobalt or ink for text on paper. Orange is an accent, paired with a label or shape. Minimum web body size: 16 px. Preserve keyboard focus, reduced-motion preferences and meaningful link text. Do not use color alone for status.

Mark rules

Clear space: at least one orange-dot diameter. Minimum mark size: 28 px or 10 mm. Use the full wordmark at 240 px or wider. Do not stretch, rotate, add effects or place another institution's name or logo beside it as a partner.

Assets in the kit

Two wordmarks, signal mark, six line icons, three campaign posters, letterhead, researcher card, email signature, report template, experiment log and web tokens. All SVGs remain editable. No third-party font or artwork is bundled.

Experiments worth trying

01 Local messages that survive a disconnect

Question. Can two local devices reconcile messages without losing a confirmed note?

One week. Build a two-client prototype with synthetic messages. Disconnect one client, create conflicting edits, reconnect and log 20 trials.

Four weeks. Add explicit conflict choices, export and a reproducible three-device test.

Skills / materials. JavaScript, storage, test design • \$0-50; existing laptops; optional local router

Data and safety. Use invented messages. Keep the demo on a trusted local network; no identity claims or production encryption claims.

Deliverable / target. Runnable demo, event log, conflict policy and two-page findings; Zero lost confirmed messages in the 20 scripted trials; all conflicts visible

Human boundary. Mentor reviews storage and threat model. Mike approves any external pilot; agents propose tests and never read personal messages.

02 Community radio listening desk

Question. How can a listening guide make a lawful public broadcast easier to discover?

One week. Compare three layouts using licensed sample audio or lawful public streams; record completion time for 10 scripted lookup tasks.

Four weeks. Build an accessible schedule browser with provenance and a small editorial workflow.

Skills / materials. Audio basics, interaction design, accessibility • \$0-80; headphones; optional receive-only equipment

Data and safety. No RF transmission, private-signal interception or rebroadcast. Check stream permissions before embedding or recording.

Deliverable / target. Receive-only listening prototype and source-permissions ledger; At least 9 of 10 lookup tasks completed; every audio source has a documented permission basis

Human boundary. Mentor approves sources and equipment. Human review is required before public audio or any recording of people.

Experiments worth trying

03 A quiet ambient status display

Question. Can a glanceable display convey useful status without becoming surveillance?

One week. Prototype three paper or browser states for shared equipment availability using synthetic events; test label comprehension internally.

Four weeks. Add an optional low-voltage e-paper display and user-controlled quiet hours.

Skills / materials. Visual design, basic electronics, prototyping • \$0-120; paper or optional USB-powered display

Data and safety. No cameras, microphones, people tracking or mains wiring. Use status and text as well as color.

Deliverable / target. State vocabulary, browser prototype and installation sketch; Four of five consenting adult testers identify each state within five seconds

Human boundary. Mentor approves any hardware and placement. Mike approves public installation; agents cannot infer people's presence.

04 Accessible message repair

Question. Which error message best helps a person recover from a failed send?

One week. Create keyboard-first send/error/retry flows with screen-reader labels. Run 12 scripted keyboard and announcement checks.

Four weeks. Co-design revisions with consenting adult participants under an approved accessibility test plan.

Skills / materials. HTML, CSS, accessibility testing, plain language • \$0-40; existing laptop; free assistive technology

Data and safety. Do not infer disability or collect medical data. Participation is optional; record only necessary anonymous observations.

Deliverable / target. Accessible prototype and keyboard/screen-reader test sheet; All 12 checks pass; failures remain understandable without color or sound

Human boundary. Mentor signs off test design and consent. Human approval precedes participant recruitment and recording.

Experiments worth trying

05 An agent handoff you can inspect

Question. What must a handoff contain so another agent can continue without inventing authority?

One week. Define a handoff schema: task, evidence, permissions, unknowns and stop condition. Compare it against free text on 20 synthetic tasks.

Four weeks. Add validation, replay traces and a human approval checkpoint for external actions.

Skills / materials. JSON, evaluation design, prompt engineering • \$0-60; optional capped model API usage

Data and safety. Use synthetic tasks and redact secrets. No autonomous sending, purchases or permission escalation.

Deliverable / target. Schema, validation tests and scored evaluation set; Every blocked action stays blocked; at least 18 of 20 handoffs preserve the stated next step

Human boundary. Mentor reviews rubric and API cap. Humans approve consequential actions; agents can draft and test within the sandbox.

06 Offline field notes that come home

Question. How can notes remain usable offline and merge predictably later?

One week. Build a local notebook with timestamps and export. Run 15 airplane-mode/edit/reconnect scenarios using invented field observations.

Four weeks. Add conflict comparison and a portable archive with attachment checksums.

Skills / materials. Web storage, file formats, UX testing • \$0-40; existing laptop or phone

Data and safety. No real location trails, private field observations or participant identifiers. Test deletion and explain local-device exposure.

Deliverable / target. Offline demo, export specimen and recovery instructions; All 15 scenarios preserve original text; a second machine opens the export

Human boundary. Mentor reviews data lifecycle and merge rules. Any real field study requires human approval and a data plan.

Experiments worth trying

07 Public data with visible uncertainty

Question. Can a small dashboard explain stale or missing public sensor readings honestly?

One week. Choose one appropriately licensed public dataset; cache a snapshot and prototype current, stale, missing and suspect states.

Four weeks. Compare two aggregation methods and publish reproducible transformations with attribution.

Skills / materials. Data cleaning, chart design, basic statistics • \$0-30; public dataset; existing laptop

Data and safety. Use public non-personal data only. No location inference about people and no emergency, health or safety guidance.

Deliverable / target. Dashboard prototype, license record and transformation notebook; Every chart shows unit, source and timestamp; all four failure states are visibly distinct

Human boundary. Mentor approves dataset and uncertainty language. Mike approves publication; agents may summarize but cannot certify accuracy.

08 Translation with a visible review step

Question. Where do short community notices lose meaning during translation?

One week. Write 20 invented notices; compare draft translations with qualified human review for names, dates, instructions and ambiguity.

Four weeks. Add a glossary, side-by-side editor and disagreement log for two reviewed languages.

Skills / materials. Language skills, content design, evaluation • \$0-40 materials; human review must be separately budgeted

Data and safety. No sensitive notices, medical/legal advice or real personal data. Do not label machine output as human-verified.

Deliverable / target. Reviewable prototype, glossary and error taxonomy; All names, dates and instructions match after human review; uncertainty remains visible

Human boundary. A qualified human reviewer is required for each language. Mike approves any public translation; agents only draft.

Experiments worth trying

09 A small archive you can reopen

Question. What is the simplest durable record of a research experiment?

One week. Package one synthetic experiment as README, open-format data, assets and checksums. Ask a second person to follow it on a clean folder.

Four weeks. Add versioned manifests, consent/license fields and a migration checklist.

Skills / materials. Documentation, file organization, basic scripting • \$0-30; existing storage

Data and safety. No proprietary work, participant identities or unlicensed media. State retention and deletion limits before real archiving.

Deliverable / target. Portable archive, manifest schema and recovery report; A second person locates evidence and reproduces the sample result without author help

Human boundary. Mentor approves licensing and retention. Human approval is required before archiving real personal or third-party records.

10 Useful interfaces on a thin connection

Question. Which content is truly necessary when a connection is slow or unreliable?

One week. Create a text-first notice page; simulate 400 kbps and 400 ms latency with cache disabled. Compare against a heavier baseline.

Four weeks. Add offline fallback, resilient form-free navigation and a documented asset budget.

Skills / materials. Frontend performance, HTML, accessibility • \$0-40; browser dev tools; existing laptop

Data and safety. Use synthetic content and local network simulation. No covert analytics or live user profiling.

Deliverable / target. Working prototype, trace captures and byte-budget report; Core text usable within five seconds in 9 of 10 controlled runs; no broken keyboard navigation

Human boundary. Mentor validates measurement setup. Mike approves deployment and any analytics; agents may run synthetic checks only.

Experiments worth trying

11 PointCast community bulletin

Question. Can a shared bulletin make a small place easier to navigate without gathering personal data?

One week. Mock a PointCast-area bulletin with five invented notices, expiry dates and clear source labels. Run 10 find-and-verify tasks.

Four weeks. Build moderation, expiry and correction views; explore a print companion and local cache.

Skills / materials. Service design, frontend, content governance • \$0-70; printing or optional display mockup

Data and safety. PointCast is a program context, not a verified venue partnership. No real names, addresses, contact forms or live submissions.

Deliverable / target. Bulletin prototype, moderation checklist and print sheet; At least 9 of 10 tasks find the right notice; every notice shows source and expiry

Human boundary. Mike confirms the place, partners and permissions before any installation. Human moderation approves every real public notice.

12 The reproducibility relay

Question. Can a new teammate repeat a result from the record alone?

One week. Take another project's synthetic dataset and instructions. Re-run three key tests independently and document discrepancies.

Four weeks. Build a reusable evidence checklist, comparison report and release gate across four projects.

Skills / materials. QA, documentation, skeptical reasoning • \$0-40; existing laptop

Data and safety. Only use approved synthetic/public data. Do not copy secrets into logs or let agents mark their own work independently verified.

Deliverable / target. Independent replication note, annotated checklist and corrected instructions; Three tests reproduce within declared tolerance or yield an explicit, actionable failure report

Human boundary. A human mentor defines tolerances and reviews discrepancies. Mike decides release; agents help execute but do not approve themselves.

People and agents working together

Mike

Proposed sponsor and scope owner; confirms supervisor, budget, partnerships and release authority before launch.

Interns

Proposed supervised researchers; scope, pay and working terms must be settled before any work begins.

Agents

Assist with research, synthetic data, code, tests and documentation; cannot replace human supervision, consent or approval.

A proposed sequence

Before launch: confirm pay, hours, supervisor, eligibility, location and legal terms.

2026 proposal: choose one or two low-risk pilots, finalize scope and prepare safe fixtures.

One-week experiment: question, baseline, small build, repeated test, evidence review.

Four-week stretch: refine, independently reproduce, document and seek release approval.

2027 proposal: revisit successful pilots with mentors; publish only verified results and limitations.

The release gate

Before sharing a prototype: review permissions, source licenses, accessibility, retention, security and unsupported claims. Publish the method, result and limitation together. A human reviewer signs the release; passing a synthetic test does not prove production readiness.

A place to test an idea carefully

PointCast is an exploratory program context; exact place, partner and installation permissions remain to be confirmed.

Start with a bulletin prototype

Project 11 gives the PointCast area a concrete starting point: a locally legible bulletin, a print companion and an expiry policy. Begin with invented notices and paper or browser mockups. Avoid implying that a real venue, installation or community partnership already exists.

Questions before a real pilot

Who owns the site? Who can approve a display? Who checks notices, corrects errors and removes expired material? Which languages and access needs matter? Can the experiment work without collecting personal data?

Consent and data limits

Use synthetic or properly licensed public data by default. Human-subject participation needs an approved plan, voluntary consent, minimal collection, deletion rules and the right to stop. Do not recruit children, collect health data or install recording devices through this program kit.

What counts as a useful result

A small negative result can be valuable if the method is clear. A report should distinguish what was observed, what is inferred and what remains unknown. The lab does not claim a breakthrough, product certification or invention simply because a prototype runs.

Materials and work terms

Pay, hours, eligibility, supervisor, location, agreements and legal review must be resolved before recruitment or work begins. Budgets are illustrative materials estimates in USD, exclude labor, tax and shipping, and are not spending authorization.

Leave a record someone can use

The one-page research note

Use the editable template to record: question; scope and authority; method; evidence; what failed; conclusion; reproduction steps; and release check. Name a human reviewer and retain artifact references. The experiment log stores expected and observed results separately.

Campaign and stationery

The three posters introduce useful signals, inspectable handoffs and open questions. They announce a proposed research program, not recruitment. The researcher card contains placeholders until a role and contact are confirmed. Do not invent qualifications, contact information or partners.

Implementation checklist

Use content.json for project pages. Preserve the proposed-program status, affiliation disclaimer and budget caveat. Keep project detail readable on mobile and offline where possible. Do not add application forms or analytics without a separate approved plan.

Sources and historical context

[1] Nokia Bell Labs history. Historical context for the transistor and information theory; accessed 2 October 2026.
<https://www.nokia.com/bell-labs/about/history/>

[2] Nokia Bell Labs research areas. Context for cross-disciplinary research. The principles and identity in this kit are original proposals, not Bell Labs policies.
<https://www.nokia.com/bell-labs/research/>

Scope of this guide

Identity, project briefs, templates and operating proposals are original work for this program. Historical references provide context only. No Bell Labs logo, affiliation, endorsement, patent claim or inherited research authority is asserted.